Equal Pixels, Unequal Views: Field of View Limits Robust AI-Generated Image Detection

Anonymous submission

Abstract

AI-generated image detectors commonly score fixed-size crops taken at native resolution, following the forensic advice to avoid resizing. A fixed-size crop shows less of a larger image, so on large images the detector sees far less of the scene than it did in training. We show that giving a frozen, crop-trained DINOv2 detector the whole image instead improves robustness with no retraining. On the hard validation split of the NTIRE 2026 Robust AI-Generated Image Detection challenge, robust AUC rises from 0.886 to 0.963. On large degraded images from the held-out AIGIBench SocialRF benchmark, it rises by 8 to 9 points. Controlled experiments show that the gain comes from a wider field of view in each input, the share of the image one input shows, and not from tiling more of the image with extra crops at the same scale. Downscaling the image before cropping widens each crop and helps DINOv2 on both datasets, and CLIP and the public detector B-Free on SocialRF. Stronger downscales may also weaken the degradation itself. The gain grows with degradation strength and image size, while on large clean images whole-image input lowers clean AUC. Every test was pre-registered. We also report two label leaks in public benchmarks.

1 Introduction

Deployed detectors of AI-generated images see what social platforms deliver: images that have been recompressed, resized and filtered, at every size. A detector must turn each one into a fixed-size input. Forensic work argues that this step should not resize: "Most of all, it is important to avoid resizing" because interpolation "may erase the subtle high-frequency traces left by the generation process" (Corvi et al. 2023). The usual answer is a fixed-size crop at native resolution, which we call a *window*, scored once or several times with the scores averaged (Corvi et al. 2023; Li et al. 2025a; Guillaro et al. 2025; Konstantinidou, Koutlis, and Papadopoulos 2025).

A fixed-size window keeps pixel count equal across images. It does not keep field of view equal. We call each square input the model scores a *view*, and define its field of view as the share of the image's short side it shows: a window of side s on an image with short side h has FOV $= s/h$. A 200-pixel window covers 39% of the short side of a 512-pixel image and 5% of a 4000-pixel one, so the larger the image, the less of it each view shows. FixRes (Touvron et al. 2019) describes a related effect in image classification, where a change in apparent object size between training and test costs accuracy ("networks do not have built-in scale invariance"). That account, a covariate shift (Shimodaira 2000) in the input, predicts that a detector does best when the test view matches the training view.

Our detector fits this account. We built it for the NTIRE 2026 Robust AI-Generated Image Detection challenge (Gushchin et al. 2026). All 407k training images have a short side of at most 512 pixels, so a 200-pixel window at native scale showed at least 39% of the short side. Augmentation could crop a window further, to no less than 20%, and always enlarged what remained. On the organisers' hard validation split a native window shows as little as 5%, and robust AUC fell as native image size grew. Three pre-registered method changes aimed at the degradations did not help. Showing the whole image did, by 7 to 9 points per branch, with no retraining. The limit is the field of view of each view. Pixel count matters little, and so does *coverage*, the share of the image that all views show together when we tile it with more windows of the same size.

Contributions.

- A diagnosis. For our vision transformer detectors on degraded images, the field of view of each view limits robust AUC, and pixel count does not. Whole-image input recovers 7 to 9 points per branch with no retraining, and the gain is not a resolution shortcut (Secs. 4 and 5).
- A mechanism. Controlled experiments separate field of view, coverage and degradation removal. The gain comes from a wider field of view in each view, not from tiling more of the image with same-size windows. A 2x downscale removes no measurable degradation for DINOv2 and a little for CLIP, and stronger downscales may also reduce it. The gain grows with degradation level and image size (Sec. 6).
- Reach and baselines. The downscale gain holds for DINOv2, CLIP and B-Free, a public detector built on native-resolution traces. Whole-image input beats three alternatives: a native window with no enlargement, multi-

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

scale training, and a learned combiner over nine windows (Secs. 7 and 8).

- Where it stops. On large clean images whole-image input costs clean AUC, for B-Free as for our detectors, and we cannot yet say why. Every detector we tested is a vision transformer. Along the way we found two label leaks in public benchmarks: a file-format and size confound in AIGIBench CommunityAI (Li et al. 2025b), and a non-monotone size leak in the resized RAISE reals commonly paired with Synthbuster (Bammey 2024) (Sec. 9).

Whole-image input is not new. The NTIRE 2026 winner resizes the whole image at inference "to preserve the global spatial context", the fifth-placed team calls its preprocessing "squish" (Gushchin et al. 2026), and AI-GenBench (Pellegrini et al. 2025) finds resize "often preferable" to multi-crop for CLIP and DINOv2 backbones. Each gives a one-line rationale: AI-GenBench notes that multi-crop uses "only partial content information", and the NTIRE entries cite global context or full image content. None separates each view's field of view from coverage, pixel count and degradation removal, or tests the rationale. Our contribution is that test and the evidence around it. We also show where the advice to avoid resizing still holds.

2 Related work

Crop, don't resize. Corvi et al. (2023) recommend training on native patches cropped without resizing and deciding on the whole image by fusion. SAFE (Li et al. 2025a) replaces down-sampling with a crop and reports that the resize "indeed diminishes the subtle artifact features". B-Free (Guillaro et al. 2025) trains on 504-pixel crops and averages native-resolution crops at test time. SPAI (Karageorgiou et al. 2025) learns a spectral model so that any resolution can be scored without resizing. TextureCrop (Konstantinidou, Koutlis, and Papadopoulos 2025) selects textured native crops and notes that preprocessing loss grows with native resolution. HiDA-Net (Mu et al. 2026) tiles the full image and reports 84.2% accuracy from its global view alone against 96.1% with tiles. All of these reason about high-frequency traces. SAFE's own ablation is useful context for us: its CNN scores a centre crop and the full native image about equally, so a low-level detector does not appear to depend on field of view.

Resize can win. AI-GenBench (Pellegrini et al. 2025) reports resize at least as good as five-crop for fine-tuned DINOv2 ViT-L/14 (94.24 against 93.44 AUROC) and clearly better for CLIP (92.04 against 87.66). In AIGIBench's test-time comparison (Li et al. 2025b), the one CLIP-based detector does better with resize while all five low-level detectors do better with crops. Cozzolino et al. (2024) resize the short side before a centre crop.[1] WaRPAD (Choi, Lee, and Lee 2025) rescales the whole image to a fixed square before tiling it, which holds each tile's field of view constant, although the paper motivates it differently. GlobalForge (Cui et al. 2026) reports the opposite of our test-time result on clean data: a UnivFD probe trained with centre crops loses about 9 balanced-accuracy points (67.9 to 58.7, their Fig. 2) when tested with resize. We return to this in Sec. 8.

[1] In their released code, https://github.com/grip-unina/ClipBased-SyntheticImageDetection.

Scale and size as confounds. Grommelt et al. (2025) show that JPEG quality and image size separate real from fake in GenImage, and note in one sentence that "detectors employing cropping to achieve a uniform input size are still subject to bias in terms of object size". Their concern is a size shortcut correlated with the label. Ours is the reverse case: size is uninformative in training and the detector still fails across sizes at test time. Rajan et al. (2025) show that random resized crops inject resampling signals when real and fake sizes differ, and BIAS-ID (Ricker, Fischer, and Quiring 2026) measures detector score shifts under such transformations. FixRes (Touvron et al. 2019) is the closest precedent. Sec. 6 shows that its train-test account fits our data at the level of a single view.

3 Setup

Data. We train on 407,223 images: the NTIRE 2026 training shards, COCO and Open Images reals, and paired real and generated images from several open and commercial generators. Every training image has a short side of 200 to 512 pixels (median 512, and 384 or more for 95% of images). We evaluate on the organisers' labelled splits: $\mathrm{ntire_val}$ (10,000 images, half distorted), $\mathrm{ntire_val_hard}$ (2,500, half distorted) and the public test split $\mathrm{ntire_test}$ (2,500). Robust AUC is the ROC AUC over distorted images, and clean AUC is over undistorted ones, following the challenge (Gushchin et al. 2026).

Models. Two branches built on DINOv2-L/14 with registers (Oquab et al. 2024; Darcet et al. 2024), all 24 blocks fine-tuned, mean-pooled patch tokens, 0.91B parameters in total. Branch A is one tower at a 512-pixel input, trained for 2 epochs. Branch B concatenates two towers at 224 pixels, trained for 3 epochs. Each source image yields one clean view and ten degraded views drawn from a pool of 26 operations in six families, chained up to five deep, including a neural codec (Ballé et al. 2018) and a downscale applied before the window is taken. The loss is class-balanced cross-entropy plus a degradation-prediction term (weight 0.3) and clean-degraded consistency terms on the outputs (KL) and hidden features (MSE), each at unit weight. The schedule is cosine with 3% warm-up, with weight averaging (Izmailov et al. 2018) over the last quarter. On one RTX 4090 the squish-trained branches took about 19 hours (B) and 29 hours (A).

Views, field of view and coverage. A *view* is one square input to the model: a region of the image resized to the branch's input size. A detector may score several views of one image and average their logits. The *field of view* of a view is the share of the image's short side it shows. *Coverage* is the share of the image area that all views of one image show together. A 200-pixel window cut after the image is downscaled by a factor k has FOV $= 200k/h$. Downscaling therefore widens every view and leaves coverage unchanged, while tiling more of the image with same-size windows raises coverage and leaves each view unchanged.

The *crop* policy takes one 200-pixel window at native resolution (the centre window at test time) and resizes it to the branch's input size. Images with a short side under 200 pixels are taken whole. The *squish* policy resizes the whole image to the input size, discarding aspect ratio: area interpolation on the short side down to the input size when the image is larger, then area interpolation to the square. Both policies give every image the same number of input pixels. Under squish, field of view and coverage are both 1. To vary the two one at a time, *tiles*(k, f) takes the centred region holding a fraction f of the image area, downscales it by k with area interpolation, covers it with 200-pixel windows, resizes each window to the input size and averages the logits. f sets coverage and k sets each view's field of view.

Fusion and selection. The two branches' logits are z-scored and combined as $wz_A + (1-w)z_B$. The z-score parameters and the weight w are fitted on ntire_val alone, by the mean of robust and clean AUC. For the squish models the choice between final and averaged weights is made the same way. For the crop models we kept the choice made earlier on both validation splits. The fitted weights were $w = 0.50$ for crop and $w = 0.55$ for squish.

Statistics. All intervals are 95% percentile intervals from 2,000 paired bootstrap resamples, stratified by label and distortion status, with the same resample indices for every model being compared.

Protocol. The test split was read once with the crop fusion (2026-10-04). We then committed a pre-registration of the squish study, which allowed exactly one more read of the test split, before training any squish model. Every later test in this paper was registered before any of its numbers existed, and none read the test split again. The supplement lists every registered test, its outcome and where it is reported, including those that failed or could not be identified. We mark the few numbers that were not registered as exploratory. We chose the squish policy after seeing its effect on ntire_val_hard, so its numbers on that split are optimistic, and we label them *selection split*. The external benchmarks and the test read carry the held-out evidence. Code, the pre-registration and per-image scores will be released with the paper.

4 Equal pixels, unequal views

Table 1 shows how much of each image the window covers. In training a native-scale window shows 39 to 52% of the short side. Augmentation widens some views and narrows others, but never below 20%. On ntire_val_hard the median is 30%, and one image in twenty gets 11% or less. Under the crop policy, fused robust AUC on ntire_val_hard falls from 0.911 for images with a short side of 80 to 336 pixels to 0.865 for images above 752 pixels. Figure 1 shows the same fall in eight equal-count bins.

To separate field of view from pixel count, we scored the two frozen crop-trained branches under four test-time presentations (Table 2). *grid9* averages nine 200-pixel native windows, nine times the pixels at the same field of view. *fov3* first downscales the short side to 600 pixels and then takes the nine windows, which raises the field of view to $1/3$. *squish* shows the whole image once.

split	n	p5	p50	p95
train, native	407,223	0.391	0.391	0.521
train, window taken	220,000	0.391	0.468	1.000
train, after second crop	220,000	0.296	0.415	0.942
ntire_val	10,000	0.100	0.312	0.893
ntire_val_hard	2,500	0.112	0.298	0.784

Table 1: Field of view of the 200-pixel window (200/short side). *Window taken* is the window as the trainer cuts it, on 20,000 seeded training images with 11 views each. A downscale before the window widens the field of view: 24.9% of views exceed 0.521 and 6.3% show the whole short side. A crop after the window narrows 18% of views below 0.391, to no less than 0.195, and enlarges them back. On ntire_val_hard, 70.8% of images fall below 0.391, and the smallest views (0.05) are about 4 times narrower than any training view.

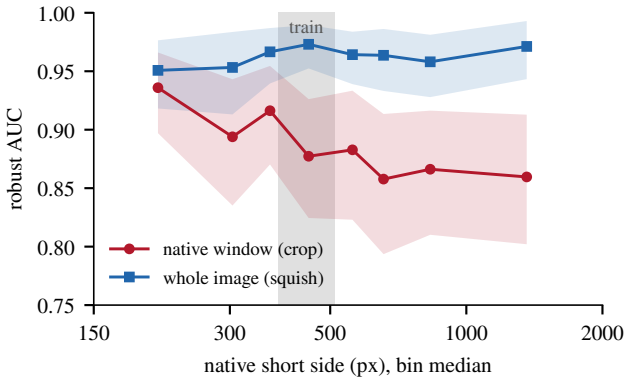


Figure 1: Fused robust AUC on the distorted rows of ntire_val_hard (selection split) in eight equal-count bins of native short side, with 95% bootstrap bands. Red: the crop-trained fusion under its 200-pixel window. Blue: the fusion trained under squish. Grey: the native short sides of the training images.

More pixels at the same field of view do little: grid9 gains 1.8 robust points on B and loses 0.7 on A. Gains rise with field of view: fov3 gains 1.4 and 3.4 points, and squish gains 9.2 and 7.3. Squish uses one view and the same pixel count as the centre window. On B the paired interval for the squish gain is [+0.057, +0.090]. With z-scores and weight fitted on ntire_val, the fused robust AUC rises from 0.886 to 0.963 and the true positive rate at 1% false positive rate from 0.516 to 0.752. Clean AUC moves by -0.001 on A and $+0.009$ on B.

Squish changes three things at once. It raises coverage to the whole image, widens the field of view of its one view, and averages any degradation over more pixels. Sec. 6 separates the three, after we rule out the simplest explanation.

branch	presentation	views	robust	clean
A (512)	centre window	1	0.8561	0.9912
	grid9	8.9	0.8495	0.9978
	fov3	8.9	0.8703	0.9977
	squish	1	**0.9482**	0.9906
B (224)	centre window	1	0.8872	0.9801
	grid9	8.9	0.9049	0.9931
	fov3	8.9	0.9215	0.9942
	squish	1	**0.9599**	0.9886

Table 2: Frozen crop-trained branches on ntire_val_hard (fit split) under test-time presentations. More pixels at the same field of view (grid9) help B by 1.8 points and cost A 0.7. Raising the field of view helps in proportion.

5 It is not a resolution shortcut

Squish does not hide native resolution. A small image arrives upsampled and soft, a large one downsampled and sharp, so a model could read size from the pixels. Four checks argue against this. Native short side scores an AUC of only 0.528 against the label on the training set, so the training data does not reward reading size. Within equal-count deciles of native short side, where the resize factor is nearly constant, squish beats the centre window in 10 of 10 deciles on branch B (size-weighted robust AUC 0.888 to 0.964, against 7.3 points pooled). When we band-limit the squished input to 160 pixels, so native resolution can no longer be read from it, branch B keeps +7.2 of its +7.3 points. Branch A keeps +5.7 of +9.2, the expected cost of removing real detail from a 512-pixel input. Last, the challenge's validation splits add crops, downscaling and perspective transforms that change native size (Gushchin et al. 2026). On rows that received only photometric transforms (median short side 784 pixels) the gain is +9.0 points on A and +8.3 on B, with intervals clear of zero.

Squish also stretches non-square images, which can introduce frequency artefacts along one axis (Grommelt et al. 2025). Two aspect-preserving presentations, the centre square after resizing the short side and a padded letterbox, keep 93 to 100% of squish's gain on both branches. Showing the whole short side drives the gain, not the stretch.

6 What the gain is made of

This section uses the frozen crop-trained branches throughout.

Degradation and size. We took the 6,250 undistorted images of the two validation splits and applied JPEG (quality 90, 70, 30), Gaussian blur (σ 0.5, 1, 2) or Gaussian noise (σ 0.02, 0.05, 0.10) at native resolution. The robust gain of squish over the centre window, averaged over the three operations, rises from -0.001 with no degradation to $+0.029$ at the heaviest level on branch A, and from $+0.004$ to $+0.027$ on B. Every rise from none to heavy has an interval above zero, pooled and in every size bucket. The rise is largest above 752 pixels ($+0.049$ on both branches). On undistorted images up to 736 pixels the two policies differ by at most 0.002. The levels are not matched in strength across operations, so we do not rank the operations.

Coverage against downscale. The images used here have a short side of at least 800 pixels. We scored *tiles*(k, f) at $k \in \{1, 2\}$ and $f \in \{0.25, 0.5, 1\}$, so every window is a 200-pixel window like those in training. The coverage effect d_F is the gain from $f = 0.25$ to $f = 1$ averaged over k. The downscale effect d_K is the gain from $k = 1$ to $k = 2$ averaged over f. On NTIRE (4,068 images, 1,235 distorted, selection split) both effects are above zero on both branches (Table 3). On large degraded SocialRF images, which no model was trained or selected on, coverage does nothing: at a fixed downscale, showing the whole image instead of its central quarter changes robust AUC by less than 0.01. The downscale helps by 2.3 to 5.5 points at every coverage level, more than on NTIRE. Covering the whole image after a 4x downscale, *tiles*$(4, 1)$, is within 0.003 of squish on NTIRE and at least as good on SocialRF (A 0.973 against 0.958, B 0.966 against 0.962). Figure 2 shows all cells: the lines are nearly flat in coverage, and downscaling lifts them.

Field of view, not degradation removal. A downscale by 2 does two things. It averages neighbouring pixels, which can remove degradation, and it doubles each window's field of view. To split them we downscaled each region by 2 and enlarged it back before cutting the windows. These windows have the field of view of *tiles*$(1, f)$ and the averaged pixels of *tiles*$(2, f)$. The difference from *tiles*$(1, f)$ is the degradation-removal part d_A, and the rest of d_K is the field-of-view part d_V. On both datasets and both branches d_A is at most 0.004 in size, with an interval that includes zero, and d_V carries the whole effect. The round-trip windows are about four times as many per image as those of *tiles*$(2, f)$, so part of d_V may come from mean pooling diluting fewer strong windows. This test cannot separate the two.

Stronger downscales. Squish downscales large images 5 to 18 times, far more than 2. To look at stronger factors we took 2,833 undistorted NTIRE images with a short side of at least 800 pixels and applied heavy JPEG (quality 30), blur ($\sigma = 2$) or noise ($\sigma = 0.10$) either before or after downscaling by k, then scored *tiles*$(k, 1)$. $R(k)$ is robust AUC with the degradation applied before the downscale minus after it, so a positive R means the downscale weakened the degradation. On branch A, $R(2) = +0.022$ [+0.019, +0.026] and $R(4) = +0.036$. On B they are +0.007 [+0.005, +0.008] and +0.013. $R(4)$ exceeds $R(2)$ on both branches, and noise, the degradation a downscale should average away most easily, gives the largest R on both (A: +0.030 at 2x, +0.047 at 4x). This fits a downscale that also reduces the degradation, more so at stronger factors. It is not proof. The same degradation may hurt more when applied to a smaller image, JPEG blocks shrink with the image, and at 2x the round trip above found no removal. We claim only that stronger downscales may also reduce the degradation.

A mismatch in each view. No training view showed less than 20% of the short side, and none at native scale less than 39% (Table 1), while a 200-pixel window on a 3,000-pixel photo shows 7%. Downscaling moves each view's field of view back toward the trained range, and squish sets it to 1, inside that range. This is a train-test mismatch in each view's

data	branch	d_F	d_K	d_A	d_V	$R(4)$
NTIRE	A	+0.005*	+0.014*	-0.002	+0.016*	+0.036*
	B	+0.011*	+0.008*	-0.001	+0.010*	+0.013*
SocialRF	A	-0.004	+0.050*	+0.001	+0.049*	n/a
	B	+0.001	+0.027*	+0.004	+0.023*	n/a

Table 3: Decomposition of the robust gain, frozen crop-trained branches, images with a short side of at least 800 pixels. d_F: coverage, from 25% to 100% of the image. d_K: downscale by 2. d_A and d_V split d_K into degradation removal and field of view ($d_A + d_V = d_K$). $R(4)$: robust AUC with heavy degradation applied before a 4x downscale minus after it, on 2,833 undistorted NTIRE images. NTIRE rows are from the selection split. SocialRF rows use our degradation chains. * marks a 95% interval that excludes zero.

branch	model	robust	clean
A	crop-trained	0.8561	0.9913
	crop-trained, squish at test	0.9482	0.9906
	squish-trained	0.9425	0.9934
B	crop-trained	0.8873	0.9801
	crop-trained, squish at test	0.9599	0.9886
	squish-trained	0.9589	0.9898

Table 4: Per branch on $\mathrm{ntire_val_hard}$ (selection split). Training under squish adds nothing to robust AUC beyond presenting the whole image to the crop-trained model. Paired gains over crop-trained: A +0.092 and +0.086, B +0.073 and +0.072, all intervals clear of zero. The crop-trained rows were rescored for this table and differ from Table 2 by at most 0.0001.

field of view, as FixRes describes, not in the image as a whole. It is not the whole story. In the bin of Figure 1 whose short sides (416 to 496 pixels) lie inside the training range, the crop fusion still scores 0.877 against 0.973 for squish. Many of these images were downscaled from larger originals, so objects in them may appear at a different size than in training.

CLIP. We repeated the decomposition on three short CLIP ViT-L/14 runs trained on crops (Sec. 7), on the same SocialRF images. Downscaling helps by 5.4 to 7.6 points and coverage does nothing, as on DINOv2. Most of the gain is the field-of-view part (d_V +0.041 to +0.063), but on CLIP the round trip also removes a little degradation (d_A +0.011 to +0.014, above zero on every seed). No measurable removal at 2x is therefore a DINOv2 result.

7 Training and baselines

Training under whole-image input. In short training runs (one tower, one epoch on a quarter of the training set, three seeds), squish-trained models beat crop-trained ones by +0.083 [+0.062, +0.105], +0.093 [+0.072, +0.115] and +0.078 [+0.057, +0.102] robust AUC on $\mathrm{ntire_val_hard}$, with clean AUC within noise. We then trained the full branches under squish (Table 4). They match the test-time change on robust AUC (A: 0.943 against 0.948, B: 0.959 against 0.960) and add a little clean AUC. The fused squish-trained models reach 0.960 robust and 0.995 clean, against 0.886 and 0.990 for the crop fusion: +0.074 [+0.058, +0.091]. The size gradient is gone: across the eight bins of Figure 1 the squish fusion stays between 0.951 and 0.973. In the short runs, training under squish adds 4 to 6 points beyond the test-time change, with DINOv2 and with CLIP (an exploratory comparison). The full branches show no such gap. They differ from the short runs in training length, number of views per image and loss terms, and we cannot say which of these closes the gap.

A second backbone. We repeated the short-run comparison with a CLIP ViT-L/14 tower (Radford et al. 2021) in place of DINOv2, every other setting unchanged. Over three seeds, training under squish beat crop by +0.139 to +0.150 robust AUC, with clean AUC unchanged, and showing the crop-trained checkpoints the whole image gained +0.094 to +0.111 at test time alone, every interval clear of zero. Both gains grow with size: at test time, +0.168 above 752 pixels against +0.057 below 336. CLIP's crop-trained short runs are weaker than DINOv2's (0.748 robust against 0.799) and gain more.

Is the crop baseline fair? The crop policy enlarges its 200-pixel window to the input size, 1.12 times for B and 2.59 times for A, and averaging native windows may dilute one informative window. We checked both. A 224-pixel native window fed at 224, with no enlargement, scores within 0.013 of the 200-pixel window, and the squish-trained short runs beat it by +0.077 on average over three seeds. At 512 pixels a native 512-pixel window beats the enlarged 200-pixel one by +0.042 [+0.025, +0.058] at one seed, but it also has 2.56 times the field of view. A 196-pixel window fed at 196 pixels has the field of view of the enlarged 200-pixel window without the enlargement, and separates the two. On 630 large degraded images the native 512-pixel window leads by 0.059. The field-of-view part is +0.048 [+0.022, +0.072]. The enlargement part is +0.011 [-0.014, +0.037], too uncertain to call, so whether enlargement handicaps branch A's crop baseline remains open. Multi-scale training with random resized crops (area 0.20 to 1.00) closes 73 to 81% of the gap between crop and squish, and squish stays ahead by +0.019 on average, every interval clear of zero. On the full frozen branches, a learned attention combiner over the nine native windows, fitted on $\mathrm{ntire_val}$, gains about 3 points over averaging them. Squish, which is not fitted, still beats it by +0.069 [+0.052, +0.087] on A and +0.023 [+0.011, +0.036] on B, and ties it on B for images below 336 pixels.

8 Held-out evidence and where it stops

External benchmarks. We scored three external benchmarks that no model was trained or selected on, after removing any image whose content digest matched an NTIRE image: AIGIBench SocialRF and CommunityAI (Li et al. 2025b), and Synthbuster with RAISE-1k reals (Bammey 2024; Dang-Nguyen et al. 2015). All three are clean, so we

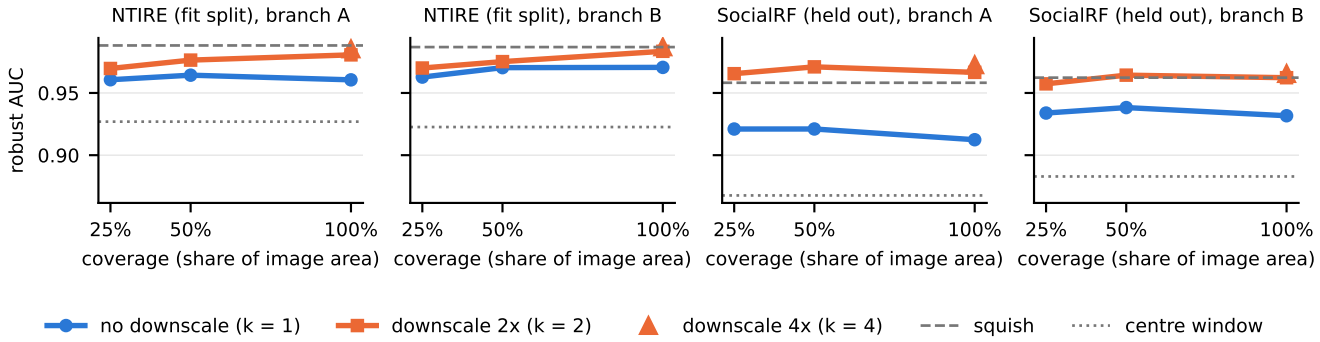


Figure 2: Robust AUC of $tiles(k, f)$ for the frozen crop-trained branches on images with a short side of at least 800 pixels. Moving right tiles more of the image with same-size windows (coverage). Moving up a line downscales the image first, widening each view. On held-out SocialRF the lines are flat in coverage and the downscale carries the gain. Dashed: squish. Dotted: the centre window.

report pooled AUC on the files as distributed, on every file re-encoded as JPEG at quality 95, and on every file passed through one seeded chain from our degradation pool. That last reading is our own synthesis, not the organisers'.

Table 5 gives the results. On SocialRF, squish wins on the files as distributed, after JPEG re-encoding, and within size deciles, and on its large degraded images the gain comes from the downscale (Sec. 6). On CommunityAI the pooled gain is confounded: every real image is a JPEG and every fake a PNG, and native size alone separates the classes with AUC 0.916. The size-matched gain on the re-encoded files does not clear zero ([−0.002, +0.012]). On Synthbuster squish loses by 0.001, with both fusions near ceiling. Under our degradation chains squish wins on all three sets, by 2.4 to 8.5 points.

The second test read. Table 6 gives both reads of the public test split. Read 1 used the crop fusion as frozen on 2026-09-23. Read 2 used the squish fusion as frozen before any squish model was scored on test, and our pre-registration allows no third read. Robust AUC rose by +0.062 [+0.040, +0.083] and clean AUC fell by 0.014 [0.006, 0.024]. On robust AUC the squish fusion would sit between the sixth- and seventh-placed challenge entries on the open test, where the crop fusion was below all nine. We were not a challenge entry and make no ranking claim. The leading entries use far more compute, such as committees of DINOv3 models at 512 pixels on 32 A100 GPUs (Gushchin et al. 2026).

The clean cost. The clean loss on the test split has one source. Clean AUC of branch A on test images with a short side above 1100 pixels falls from 0.974 under crop to 0.880 under squish (366 images). Below 1100 pixels, and on branch B at every size, the change is under 1.3 points. The fake images drive it: their mean score under squish falls with size. The validation splits do not show this cost, and the test split carries no generator labels. Synthbuster, which has them, shows no single location either. The two A models also differ in weight averaging and training round, not only in input policy. We tried to remove the cost with a per-image mix of squish and grid9, weighted by the branch's predicted degradation. It lost robust AUC against squish alone (-0.005 [-0.010, -0.001]), so we did not spend a test read on it. Our hypothesis is the trade-off the forensic literature predicts: a clean, large generated image keeps fine traces at native resolution, and reducing it to 512 pixels discards them. This would also reconcile our result with GlobalForge (Cui et al. 2026) and TextureCrop's CLIP results (Konstantinidou, Koutlis, and Papadopoulos 2025), both measured on clean data. Once blur, compression or downscaling has damaged the traces, the detector has to judge content, and each view needs a wider field of view. We have not tested this hypothesis directly.

Public detectors. UnivFD (Ojha, Li, and Lee 2023), a linear probe on CLIP ViT-L/14 features trained on 224-pixel crops, is at chance on NTIRE data under every presentation (robust AUC 0.49 to 0.51 on the hard split), so it says nothing either way. B-Free (Guillaro et al. 2025) is a DINOv2-B with registers that averages five 504-pixel windows taken at native resolution.[2] We scored it with its authors' pipeline (native), at its own training field of view (short side to its training size, then its crop), and under squish (Table 7). Squish changes its robust AUC on the hard split by -0.018 [-0.041, +0.004], and the change falls with image size instead of rising. Both resized presentations lower its clean AUC on every split, by 1.2 to 18.8 points, as the advice to avoid resizing predicts for a detector built on native-resolution traces. Under controlled degradations its squish deficit shrinks, from -0.114 with none to -0.035 at the heaviest level, but never crosses zero. On SocialRF under our degradation chains squish gains it 6.3 points [+5.1, +7.5]. On large degraded SocialRF images (short side at least 2,016 pixels), with B-Free's own 504-pixel windows, downscaling before cutting windows raises its robust AUC by 7.9 points [+5.3, +10.5] while coverage changes it by -0.003, and squish scores 0.928 against 0.804 for its native five windows. On the clean images of the same

[2]Model size and window count from the released weights and code, https://github.com/grip-unina/B-Free.

	SocialRF	CommunityAI	Synthbuster
n (real/fake)	2994/3000	6000/6000	994/8996
short side p50	1366	500	833
size-label AUC	0.694	0.916	0.519
format-label AUC	0.500	1.000	0.555
crop fusion	0.9633	0.9809	0.9995
squish fusion	0.9776	0.9920	0.9984
difference	+0.014	+0.011	-0.001
95% CI	[+0.010, +0.018]	[+0.009, +0.013]	$[-0.002, -0.001]$
size-matched, JPEG 95	+0.016	+0.005	n/a
95% CI	[+0.009, +0.024]	$[-0.002, +0.012]$	
degraded (ours)	+0.085	+0.070	+0.024

Table 5: External benchmarks, fused AUC. Size-matched is the size-weighted AUC within deciles of native short side. It cannot be computed on Synthbuster because no decile holds both classes (Sec. 9).

read	robust	clean	AUC	TPR@1%
1, crop	0.7931	0.9856	0.9108	0.289
2, squish	0.8547	0.9715	0.9211	0.300

Table 6: The two reads of ntire_test (2,500 images). Read 1's fusion weight was fitted on both validation splits ($w = 0.45$), read 2's on ntire_val alone.

split	native	train FOV	squish
hard-val, robust	0.636	0.631	0.618
hard-val, clean	0.909	0.753	0.722
ntire_val, robust	0.735	0.754	0.744
ntire_val, clean	0.920	0.848	0.825
SocialRF	0.922	0.889	0.871
degraded (ours)	0.757	0.829	0.820
CommunityAI	0.891	0.742	0.703
degraded (ours)	0.683	0.568	0.539
Synthbuster	0.998	0.987	0.976
degraded (ours)	0.896	0.893	0.892

Table 7: B-Free AUC under three presentations. Native is the authors' own five-window test pipeline. Hard-val is ntire_val_hard.

set, downscaling costs it 1.7 points and squish 3.3. B-Free thus shows the same trade as our detectors: each view's field of view limits it on degraded images, and native resolution helps it on clean ones.

9 Two label leaks

AIGIBench CommunityAI. All 6,000 real images are JPEG files and all 6,000 fake images are PNG files, so container format separates the classes perfectly, and native short side separates them with AUC 0.916. We found no mention of this in the paper or its public issue trackers. Results on this subset should be reported after format equalisation and within size strata.

Synthbuster with resized RAISE reals. Synthbuster ships only generated images (Bammey 2024).[3] A widely used copy of its RAISE reals, released with B-Free (Guillaro et al. 2025), was resized so that each image's area is about 1024^2 pixels.[4] We used this copy. All 994 reals then have a short side of 833 to 835 pixels, while the fakes range from 256 to 2048. The monotone size-label AUC is 0.519 because the reals sit in the middle of the fakes' range, but the rule "short side within 833 to 835" separates the classes almost perfectly. A monotone leak check misses this. In the same benchmark, Adobe Firefly images saved with a .png extension are JPEG internally (Ricker, Fischer, and Quiring 2026), and 997 of our fakes are JPEG.

10 Limitations

Every detector we tested is a vision transformer: our two DINOv2 branches, three short CLIP runs and B-Free. The low-level and convolutional detectors that motivate the advice to avoid resizing may not share the effect, and SAFE's results suggest they do not (Li et al. 2025a). UnivFD is at chance on this data. The policy was chosen on ntire_val_hard, so its numbers there are optimistic. The external sets and the test read are the unbiased evidence, and they are mixed. The decomposition uses frozen NTIRE-trained models, one held-out set, and only images of at least 800 pixels. Downscaling widens each view but also leaves fewer windows to average, and we cannot tell the two apart. The order test cannot tell degradation removal apart from degradation that hurts more on a smaller image, and it stops at 4x while squish downscales large images 5 to 18 times. The 512-pixel comparisons are single short runs, and they could not settle how much enlarging the window costs. Every training image has a short side of 512 pixels or less, so narrow views at native scale were never in training. The clean cost on large test images is real, unexplained, and not addressed here. Combining a whole-image view with native windows, as HiDA-Net and GLASS do (Mu et al. 2026; Han 2026), is the obvious next step, but it would need its own held-out test.

[3]The dataset record states that the RAISE-1k images are not included: https://zenodo.org/records/10066460.

[4]B-Free README, https://github.com/grip-unina/B-Free.

11 Conclusion

A fixed-size window at native resolution fixes pixel count but not field of view: the larger the image, the smaller the share each view shows. For our detectors on degraded images, that share limited robust AUC. The gain comes from a wider field of view in each view, not from tiling more of the image with same-size windows, and stronger downscales may also weaken the degradation. This is a train-test mismatch in each view's field of view. Showing the whole image recovered 7 to 9 points per branch with no retraining, and training the full models on whole images added nothing more. The same downscale gain appears for CLIP and for B-Free, a detector built on native-resolution traces. On large clean images the change cost clean AUC, possibly the case the advice to avoid resizing was written for. Detection papers should report the field of view of each view and the downscale factor their preprocessing implies, alongside resolution and crop policy.

References

Ballé, J.; Minnen, D.; Singh, S.; Hwang, S. J.; and Johnston, N. 2018. Variational Image Compression with a Scale Hyperprior. In *International Conference on Learning Representations (ICLR)*.

Bammey, Q. 2024. Synthbuster: Towards Detection of Diffusion Model Generated Images. *IEEE Open Journal of Signal Processing*, 5: 1–9.

Choi, S.; Lee, H.; and Lee, M. 2025. Training-free Detection of AI-generated images via Cropping Robustness. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Corvi, R.; Cozzolino, D.; Zingarini, G.; Poggi, G.; Nagano, K.; and Verdoliva, L. 2023. On the Detection of Synthetic Images Generated by Diffusion Models. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5.

Cozzolino, D.; Poggi, G.; Corvi, R.; Nießner, M.; and Verdoliva, L. 2024. Raising the Bar of AI-generated Image Detection with CLIP. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 4356–4366.

Cui, M.; Liu, R.; Zou, D.; Qin, Z.; Xu, J.; Wang, Z.; Wei, J.; Zhou, H.; Liu, Y.; Wang, Y.; and Wu, S. 2026. GlobalForge: Towards Robust AI-Generated Image Detection. *arXiv preprint arXiv:2607.14684*.

Dang-Nguyen, D.-T.; Pasquini, C.; Conotter, V.; and Boato, G. 2015. RAISE: A Raw Images Dataset for Digital Image Forensics. In *Proceedings of the 6th ACM Multimedia Systems Conference (MMSys)*.

Darcet, T.; Oquab, M.; Mairal, J.; and Bojanowski, P. 2024. Vision Transformers Need Registers. In *International Conference on Learning Representations (ICLR)*.

Grommelt, P.; Weiss, L.; Pfreundt, F.-J.; and Keuper, J. 2025. Fake or JPEG? Revealing Common Biases in Generated Image Detection Datasets. In *Computer Vision – ECCV 2024 Workshops*, Lecture Notes in Computer Science, 80–95. Springer.

Guillaro, F.; Zingarini, G.; Usman, B.; Sud, A.; Cozzolino, D.; and Verdoliva, L. 2025. A Bias-Free Training Paradigm for More General AI-generated Image Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 18685–18694.

Gushchin, A.; Abud, K.; Shumitskaya, E.; Filippov, A.; Bychkov, G.; Lavrushkin, S.; Erofeev, M.; et al. 2026. NTIRE 2026 Challenge on Robust AI-Generated Image Detection in the Wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 1895–1913.

Han, L. 2026. Attention to Detail: Global-Local Attention for High-Resolution AI-Generated Image Detection. *arXiv preprint arXiv:2601.00141*.

Izmailov, P.; Podoprikhin, D.; Garipov, T.; Vetrov, D.; and Wilson, A. G. 2018. Averaging Weights Leads to Wider Optima and Better Generalization. In *Conference on Uncertainty in Artificial Intelligence (UAI)*.

Karageorgiou, D.; Papadopoulos, S.; Kompatsiaris, I.; and Gavves, E. 2025. Any-Resolution AI-Generated Image Detection by Spectral Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 18706–18717.

Konstantinidou, D.; Koutlis, C.; and Papadopoulos, S. 2025. TextureCrop: Enhancing Synthetic Image Detection through Texture-based Cropping. In *Proceedings of the Winter Conference on Applications of Computer Vision (WACV) Workshops*, 1459–1468.

Li, O.; Cai, J.; Hao, Y.; Jiang, X.; Hu, Y.; and Feng, F. 2025a. Improving Synthetic Image Detection Towards Generalization: An Image Transformation Perspective. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.

Li, Z.; Yan, J.; He, Z.; Zeng, K.; Jiang, W.; Xiong, L.; and Fu, Z. 2025b. Is Artificial Intelligence Generated Image Detection a Solved Problem? In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

Mu, L.; Hu, H.; Zou, X.; Bai, J.; and Hu, J. 2026. No Pixel Left Behind: A Detail-Preserving Architecture for Robust High-Resolution AI-Generated Image Detection. In *International Conference on Learning Representations (ICLR)*.

Ojha, U.; Li, Y.; and Lee, Y. J. 2023. Towards Universal Fake Image Detectors that Generalize Across Generative Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 24480–24489.

Oquab, M.; Darcet, T.; Moutakanni, T.; et al. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.

Pellegrini, L.; Cozzolino, D.; Pandolfini, S.; Maltoni, D.; Ferrara, M.; Verdoliva, L.; Prati, M.; and Ramilli, M. 2025. AI-GenBench: A New Ongoing Benchmark for AI-Generated Image Detection. In *International Joint Conference on Neural Networks (IJCNN)*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.;

Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, 8748–8763. PMLR.

Rajan, A. S.; Ojha, U.; Schloesser, J.; and Lee, Y. J. 2025. Aligned Datasets Improve Detection of Latent Diffusion-Generated Images. In *International Conference on Learning Representations (ICLR)*.

Ricker, J.; Fischer, A.; and Quiring, E. 2026. BIAS-ID: A Framework for Analyzing Transformation Biases in AI-Generated Image Detectors. *arXiv preprint arXiv:2605.31153*.

Shimodaira, H. 2000. Improving Predictive Inference under Covariate Shift by Weighting the Log-Likelihood Function. *Journal of Statistical Planning and Inference*, 90(2): 227–244.

Touvron, H.; Vedaldi, A.; Douze, M.; and Jégou, H. 2019. Fixing the train-test resolution discrepancy. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32.